Statistical Inference

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HW 4: Part I

- **Important Note:** This assignment consists of 15 questions, with the last 3 involving Python implementation. Out of the first 12, you only need to answer 8 questions (4 questions are optional). For the 3 Python-based questions at the end, one is also optional.
- If you get stuck on any part of the homework, just reach out in the group chat or email to the authors. We're here to help!
- Feel free to use Python for challenging computations like binomial probabilities. That said, you are still expected to show the corresponding formulas in your final report.

Problem 1: cooling performance

A semiconductor manufacturing company wants to compare the cooling performance of two new cooling systems for their processors.

- System A (High-end Prototype): Expensive and precise, tested on a small number of prototype units.
- System B (Standard Model): Cheaper, tested on the mass production line.

The data regarding processor temperature (in degrees Celsius) after one hour of heavy load is as follows:

Parameter	System A	System B
Sample Size (n)	$n_A = 10$	$n_B = 100$
Sample Mean ($\bar{X}$)	68	74
Sample Standard Deviation (S)	4	25

The project manager claims that System A provides significantly better cooling (lower temperature) than System B.

1. Write the hypotheses test.
2. Assuming that both sample sizes are small (less than 30), calculate the pooled test statistic and determine the conclusion at $\alpha = 0.05$.
3. Assuming that both two sample sizes are large (greater than 30), calculate the new unpooled statistic, and determine the conclusion. (Consider t student instead of Z and for degrees of freedom, use the conservative approximation $df \approx n_{small} - 1$).
4. are the results from Part 2 and Part 3 the same? Explain why they are the same or different. Also explain which one is correct.
5. (optional) In these circumstances, provide the exact formula for the degree of freedom and then calculate it. Also, state the conclusion of the hypothesis test.

Problem 2: Evaluating Biomedical Treatment Effects

A biomedical researcher investigates whether a newly developed compound affects blood glucose regulation. For each of 15 patients, glucose concentration is measured twice:

- once immediately before administration of the compound,
- once 48 hours after administration.

Laboratory calibration studies have established the following properties of the measurement process:

- measurement errors are multiplicative on the original scale,
- After a logarithmic transformation, measurement errors are well-modeled as additive, approximately normally distributed, and exhibiting approximately constant variance across patients,
- inter-patient variability is additive on the log scale,
- Measurements from different patients may be regarded as independent.

Based on these considerations, the logarithmic transformation is deemed appropriate for statistical analysis. The recorded values (on the logarithmic scale) are:

Patient	Before (log)	After (log)
1	4.62	4.54
2	4.58	4.49
3	4.60	4.53
4	4.65	4.57
5	4.59	4.50
6	4.63	4.56
7	4.61	4.64
8	4.56	4.47
9	4.64	4.57
10	4.60	4.51
11	4.57	4.59
12	4.62	4.53
13	4.58	4.49
14	4.63	4.66
15	4.59	4.52

1. Modeling and parameter of interest
   a. Define a single numerical quantity derived from each patient that appropriately captures the compound's effect.
   b. Clearly state the population parameter that this quantity estimates.
2. Hypothesis formulation
   a. Formulate null and alternative hypotheses that correspond to the claim that the compound reduces blood glucose levels.
   b. Explain how your hypotheses relate to the interpretation on the original (non-log) scale.
   c. Based on the information provided, justify the form of the distribution of the statistic used to test the hypotheses.
3. Inference
   a. Compute the relevant test statistic using the data provided.
   b. Obtain the p-value and state your conclusion at the 5% significance level. clearly articulating the practical meaning of the result.

4. Interval estimation
    a. Construct a 95% confidence interval for the parameter defined in part 1.
    b. Transform the interval back to the original scale and interpret it in practical terms.
5. Diagnostic reasoning
   A colleague suggests analyzing only the direction of change (increase/decrease) for each patient, arguing that this would be "more robust."
    a. Explain why this approach is statistically inferior in the present study.
    b. Identify precisely what information would be lost and why that loss materially affects the quality of inference in this context.

Problem 3: Stelazine Treatment Outcomes

Stanley and Walton (1961) ran a controlled clinical trial to investigate the effect of the drug stelazine on chronic schizophrenics. The trials were conducted on chronic schizophrenics in two closed wards. In each of the wards, the patients were divided into two groups matched for age, length of time in the hospital, and score on a behavior rating sheet. One member of each pair was given stelazine, and the other a placebo. Only the hospital pharmacist knew which member of each pair received the actual drug. The following table gives the behavioral rating scores for the patients at the beginning of the trial and after 3 months. High scores are good.

Ward A			
Stelazine		Placebo	
Before	After	Before	After
2.3	3.1	2.4	2.0
2.0	2.1	2.2	2.6
1.9	2.45	2.1	2.0
3.1	3.7	2.9	2.0
2.2	2.54	2.2	2.4
2.3	3.72	2.4	3.18
2.8	4.54	2.7	3.0
1.9	1.61	1.9	2.54
1.1	1.63	1.3	1.72

Ward B			
Stelazine		Placebo	
Before	After	Before	After
1.9	1.45	1.9	1.91
2.3	2.45	2.4	2.54
2.0	1.81	2.0	1.45
1.6	1.72	1.5	1.45
1.6	1.63	1.5	1.54
2.6	2.45	2.7	1.54
1.7	2.18	1.7	1.54

1. For each of the wards, test whether stelazine is associated with improvement in the patients' scores.
2. Test if there is any difference in improvement between the wards.

Problem 4: Multinomial Hypothesis Test

Suppose that 400 persons are chosen at random from a large population, and that each person in the sample specifies which one of five breakfast cereals she most prefers. For i = 1, . . . , 5, let p_i denote the proportion of the population that prefers cereal i, and let N_i denote the number of persons in the sample who prefer cereal i. It is desired to test the following hypotheses at the level of significance 0.01:

$$H_0 : p_1 = p_2 = ... = p_5,$$

$$H_1 : \textit{The hypothesis } H_0 \textit{ is not true.}$$

For what values of $\Sigma_{i=1}^{5} N_i^2$ would H_0 be rejected?

Problem 5: Comparative Analysis of Virologic Failure Rates

There is a clinical study that its population consisted of HIV-infected women in sub-Saharan Africa who had been given a single dose of Nevirapine (a treatment for HIV) while giving birth to prevent transmission of HIV to the infant. The study was a randomized comparison of continued treatment of a woman (after successful childbirth) with drug A and drug B. 530 women participated in the study. Twenty-four weeks after starting the study treatment, each woman was tested to determine if the HIV infection was becoming worse (an outcome called virologic failure). 26 of the 30 women treated with drug A experienced virologic failure, while 350 of the 500 women treated with the other drug experienced virologic failure.

1. Create a two-way table presenting the results of this study.
2. State appropriate hypotheses to test for difference in virologic failure rates between treatment groups.
3. Calculate the variances of the two proportions separately and perform the test without pooling.
4. Perform the pooled test and calculate the z-statistic.
5. why are the z-values in the two methods different? Under what circumstances is it appropriate to use method A (unpooled statistc) and under what circumstances is it appropriate to use method B (pooled statistics)?
6. Select the correct method (A or B) for confidence interval and then create and interpret a 90% confidence interval of the difference in treatment.
7. Do your results from the hypothesis test and the confidence interval agree? Explain your reasoning.

Problem 6: Statistical Analysis of Vaccine Efficacy

Suppose that in a large clinical trial, the number of individuals who remain uninfected after vaccination is modeled as

$$X \sim Binomial(n = 100, p),$$

where p denotes the true infection-prevention rate of the vaccine. A pharmaceutical company claims that the vaccine prevents infection in at least 70% of cases. Consider the test that rejects $H_0 : p = 0.7$ in favor of $H_A : p \neq 0.7$ for $|X - 70| > 10$.

1. Compute the significance level α of this test exactly under H_0.
2. Graph the power as a function of p and describe how the power behaves for values of p near and far from 0.7.
3. now consider that in the actual trial, 72 out of 100 vaccinated individuals remained uninfected. Perform the hypothesis test using:
   (a) Exact binomial test (use software or table if needed).
   (b) Normal approximation (with and without continuity correction). Compute and compare the corresponding p-values.
4. Assuming the true prevention rate is p=0.75:
   Compute the exact power of the test based on the rejection rule above. Interpret this power in the context of the vaccine study. Based on both the hypothesis test and the power analysis, discuss whether the company's claim is supported by the data. Make a suggestion to increase the power.
5. Suppose that company wants to determine the minimum sample size n to ensure that the test has a power of at least 80%. Find the minimum sample size and interpret it.

Problem 7: Statistical Analysis of Sex Ratio Distribution

Consider a large population of families that have exactly three children, and suppose that it is desired to test the null hypothesis H_0 that the distribution of the number of boys in each family is a binomial distribution with parameters n = 3 and p = 1/2. Suppose also that in a random sample of 128 families it is found that 26 families have no boys, 32 families have one boy, 40 families have two boys, and 30 families have three boys.

1. At what levels of significance should H_0 be rejected?
2. suppose now that it is desired to test the composite null hypothesis H_0 that the distribution of the number of boys in each family is a binomial distribution for which n = 3, and the value of p is not specified. At what levels of significance should H_0 be rejected?

Problem 8: Pearson Chi-Square Tests

Let X be a binomial random variable with n trials and probability p of success.

1. Suppose n = 100 and X = 38. Compute the Pearson chi-square statistic for testing the goodness of fit to the multinomial distribution with two cells with $H_0 : p = 0.5$.
2. What is the approximate distribution of the test statistic in part 1, under the null Hypothesis H_0.
3. What can you say about the P-value of the Pearson chi-square statistic in part 1 using the table of percentiles for chi-square random variables?
4. Consider the general case of the Pearson chi-square statistic in part 1, where the outcome X = x is kept as a variable (yet to be observed). Show that the Pearson chi-square statistic is an increasing function of $|x - \frac{n}{2}|$.
5. Suppose the rejection region of a test of H_0 is $\{X : |X - \frac{n}{2}| > k\}$ for some fixed known number k. Using the central limit theorem (CLT) as an approximation to the distribution of X, write an expression that approximates the significance level of the test for given k. (Your answer can use the cdf of $Z \sim N(0, 1)$: $\phi(z) = P(Z \leq z)$)

Problem 9: Validation of Chi-Square Statistics

Consider a two-way contingency table with three rows and three columns. Suppose that, for i = 1, 2, 3 and j = 1, 2, 3, the probability p_{ij} that an individual selected at random from a given population will be classified in the ith row and the jth column of Table.

0.15	0.09	0.06
0.15	0.09	0.06
0.20	0.12	0.08

1. Show that the rows and columns of this table are independent.
2. Generate a random sample of 300 observations from the given population using a uniform pseudo-random number generator. Select 300 pseudo-random numbers between 0 and 1 and proceed as follows: Since $p_{11} = 0.15$, classify a pseudo-random number x in the first cell if $x < 0.15$. Since $p_{11} + p_{12} = 0.24$, classify a pseudo-random number x in the second cell if $0.15 \leq x < 0.24$. Continue in this way for all nine cells.
3. Consider the 3×3 table of observed values N_{ij} generated in part 2. Pretend that the probabilities p_{ij} were unknown, and assess whether there is statistical evidence of an association between the row and column classifications.

4. If all the students in a class carry part 1, 2 and 3 independently of each other and use different pseudo-random numbers, then the different values of the statistic Q obtained by the different students should form a random sample from the χ^2 distribution with four degrees of freedom. If the values of Q for all the students in the class are available to you, test the hypothesis that these values form such a random sample.

Problem 10: Verifying Corporate Salary Claims

A company claims that the median annual salary of its employees in the engineering department is at least 120 million Toman. To verify this claim, a small random sample of 18 employees (n=18) is selected, and their annual salaries (in millions of Toman) are collected.
125, 110, 130, 118, 140, 120, 105, 135, 122, 119, 145, 150, 115, 128, 121, 138, 112, 120

1. Define the Test Statistic and Hypotheses:
   (a) Write down the statistical hypotheses
   (b) Calculate the Sign Test statistic. How do you handle observations exactly equal to 120?
2. Exact Test:
   (a) Calculate the Exact P-value using the Binomial distribution with the appropriate parameters.
   (b) Report the test result at a significance level of $\alpha = 0.05$.
3. Normal Approximation:
   (a) Calculate the Z-test statistic using the Continuity Correction
   (b) Report the approximate P-value and compare it to the exact P-value.

Problem 11: Statistical Validation of Drug Superiority

A clinical study was conducted to evaluate the efficacy of two new weight loss medications, Drug A and Drug B. A random sample of 12 patients was treated with Drug A ($n_A = 12$), and a second independent random sample of 18 patients was treated with Drug B ($n_B = 18$). The total weight loss (in kilograms) recorded for each patient over a six-month period is provided in the table below:

Drug A ($n_A = 12$)	Drug B ($n_B = 18$)
15.2, 16.7, 17.9, 19.6, 20.2, 15.9, 18.4, 16.1, 18.9, 14.8, 19.1, 17.3	9.4, 10.3, 11.2, 15.3, 11.7, 12.1, 14.7, 12.6, 13.2, 14.1, 16.2, 17.1, 17.4, 9.9, 15.8, 16.6, 10.8, 13.6

Using the Wilcoxon-Mann-Whitney Test and the Normal Approximation method (with a significance level of $\alpha = 0.05$), perform the following statistical analyses:

1. Two-Tailed Test (Test for Any Difference): Test the null hypothesis H_0 that there is no difference in the effectiveness of the two drugs against the alternative hypothesis H_1 that the distribution of weight loss for Drug A and Drug B is different.
   - Determine the U-statistic, calculate the Z-score, and state your conclusion.
2. One-Tailed Test (Superiority of Drug A): Test the null hypothesis that Drug A is no more effective than Drug B against the alternative hypothesis that Drug A is significantly more effective than Drug B (i.e., Drug A leads to greater weight loss).
   - Carefully select the appropriate U-statistic to reflect the direction of this hypothesis.
3. One-Tailed Test (Superiority of Drug B): Test the null hypothesis that Drug B is no more effective than Drug A against the alternative hypothesis that Drug B is significantly more effective than Drug A.

- Explain how the selection of the U-statistic or the sign of the Z-score changes for this directional test.

Problem 12: Comparative Analysis of Telephone Line Fault Rates

An experiment was done to test a method for reducing faults on telephone lines. Fourteen matched pairs of areas were used. The following table shows the fault rates for the control areas and for the test areas:

Test	Control
744	88
391	570
395	605
477	617
301	653
2654	2913
851	321
622	924
186	286
770	1098
641	982
1565	2346
410	615
914	519

1. Plot the differences versus the control rate and summarize what you see.
2. Calculate the mean difference, its standard deviation, and a confidence interval.
3. Calculate the median difference and a confidence interval and compare to the previous result.
4. Do you think it is more appropriate to use a t test or a nonparametric method to test whether the apparent difference between test and control could be due to chance? Why? Carry out both tests and compare.

Problem 13: Inference for Population Variance

A) The scipy.stats library does not have a direct function for a one-sample Chi-square test for variance. Write a Python function that:

1. Calculates the χ^2 statistic and the exact P-value.
2. Returns the test result (Reject/Fail to Reject).
3. Generate 10,000 samples from $N(0, \sigma^2 = 4)$.
4. Compute χ^2 statistic each time.
5. Count proportion falling in rejection rate (Type I error)
6. Repeat part 3 to part 5 with non-normal data (t-distribution or exponential).
7. Interpret the results.

B) Perform a Monte Carlo Simulation in Python:

1. Generate two random samples from the t-student distribution (which is similar to normal but has heavier tails) with low degrees of freedom (e.g., df=3). Assume equal variances for both.

2. Run the F-test on these data and record the P-value.
3. Repeat this 10,000 times.
4. In what percentage of cases is the null hypothesis (incorrectly) rejected?
5. Is this rate close to $\alpha = 0.05$ or much higher? Interpret the result.

Problem 14: Fitting Weibull Distribution to Frequency Data

We do not have access to the raw lifetime data (in hours) of an electronic component, but the manufacturer's final report presents the data as a frequency table (histogram):

Time Interval (Hours)	Observed Count (O_i)
[0, 100)	30
[100, 300)	115
[300, 600)	180
[600, 1000)	125
[1000, ∞)	50
Total	500

The hypothesis is that the component lifetimes follow a Weibull Distribution with a shape parameter k and a scale parameter λ. The Cumulative Distribution Function (CDF) of the Weibull distribution is given by $F(x) = 1 - e^{-(\frac{x}{\lambda})^k}$. Please implement the following two parts using Python:

1. Write a function to find the parameters k and λ such that the statistic between the observed and expected frequencies is minimized.
2. Clearly state the null and alternative hypotheses and after finding the optimal parameters, perform the Test at the 5% significance level.

Problem 15: Parametric and Non-parametric Inference

To simulate a real-world statistical analysis, you will first generate your own dataset using Python. This simulates collecting data from an experiment, such as comparing the effects of two training programs on performance. Use Python for data generation, statistical computations, and visualizations where specified. Assume you have access to libraries like numpy, scipy, pandas, and matplotlib.

1. Data Generation: You are studying the impact of two training programs (Program A: intensive, Program B: standard) on performance scores (e.g., test scores out of 100, where higher is better) for participants. Use Python to generate simulated data as follows:
   (a) For Program A ($n_A = 20$ participants): Generate "before" scores from a normal distribution with mean 50 and std 10. Then, generate "after" scores by adding a treatment effect: for each participant, add a random improvement from a uniform distribution between -10 and 40, but introduce skewness by multiplying the improvement by a factor from an exponential distribution (scale=1) for half the samples.
   (b) For Program B ($n_B = 25$ participants): Generate "before" scores from a normal distribution with mean 52 and std 12. Then, generate "after" scores by adding a smaller random improvement from a uniform distribution between -5 and 10, without additional skewness.
   (c) Ensure the data has no ties for simplicity in ranking tests.
   (d) Compute improvement scores for each participant.

(e) Save the data in a pandas DataFrame and print summary statistics (mean, median, min, max) for improvements in both groups to verify non-normality (e.g., use scipy.stats.shapiro for normality check).

Provide your Python code for showing the generated dataset

2. Using the improvement scores for Program A:

   (a) Perform a one-sample sign test (exact method) to test the two-tailed hypothesis the improvement of program A ($\alpha = 0.05$). Compute the test statistic and critical value. Then, implement the asymptotic one-sample sign test in Python (using binomial approximation) to find the p-value. Compare results and discuss why the asymptotic version is appropriate for n=20.

   (b) Next, apply the paired Wilcoxon signed-rank test (exact method) on the before-after pairs for Program A. Compute the signed-rank statistic manually and show at least 5 samples, then use Python to get the full statistic and p-value (two-tailed). Interpret if Program A shows significant improvement, and explain why this test is more powerful than the sign test here (name a python function for performing wilcoxon signed-rank test directly but don't use it).

3. Using the improvement scores from both groups:

   (a) Perform the two-sample Wilcoxon signed-rank test to test the null hypothesis of no difference in improvements between Program A and Program B against the one-tailed alternative that Program A has greater improvement ($\alpha = 0.05$). Compute the U statistic manually (show ranks for the first 10 combined samples), then use Python for the exact p-value (name a python function and for performing Mann-Whitney U test directly but don't use it).

   (b) Apply the asymptotic Mann-Whitney test. compute the Z-score and p-value in Python. Compare exact vs. asymptotic results, and discuss conditions under which the asymptotic version is valid.

   (c) If ties were present in your data (simulate adding one), explain how it affects the test and adjust your Python code accordingly.

4. Parametric Comparisons and Contrasts:

   (a) Assuming normality (even if your data fails the Shapiro test), perform a parametric paired t-test on Program A's before-after scores (two-tailed, $\alpha = 0.05$). Implement in Python and compare p-value to the Wilcoxon from Part 2. why might the t-test be invalid here?

   (b) For the two groups' improvements, perform a test of differences between two means (assuming unknown variances, small n but use t-distribution). Use Python for a one-tailed test. Compute critical regions and p-values. Compare p-value to Part 3.

5. Advanced Integration and Simulation Analysis:
   Simulate 1000 iterations in Python: Generate new datasets each time (same parameters), perform the asymptotic one-sample Wilcoxon signed-rank test on Program A improvements, and count how often you reject the null (power estimation). Plot a histogram of p-values. Interpret the empirical power.